

A Short Note on New Indexing Polynomials of Finite Fields

EIJI SUGIMOTO

Otaru University of Commerce, Midori 3, Otaru, Hokkaido 047, Japan

This paper contains a list of new indexing polynomials over prime fields and presents an efficient algorithm employed in the search for them.

1. INTRODUCTION

A large amount of computation on the finite field is often required for dealing with combinatorial problems. The addition and multiplication operations on a finite field can conveniently be performed by expressing its elements in terms of an indexing polynomial over the prime field. Lists of indexing polynomials have been published by Watson (1962) and by Alanen and Knuth (1964), which are extensive for the prime field of order 2.

A table of indexing polynomials over prime fields of order 3 to 47 is presented at the end of this paper; all the items in which are not in the lists cited above. The rest of the paper is devoted to a brief explanation of the algorithm employed in the search for them. The Alanen-Knuth algorithm is described, and our refined version based on a theorem of Berlekamp follows.

2. FINITE FIELDS AND INDEXING POLYNOMIALS

For every prime p and a positive integer n there exists exactly one finite field, up to isomorphisms, of order p^n , which is referred to as $GF(p^n)$. By a primitive element θ of $GF(p^n)$ is meant a multiplicative generator of $GF(p^n) - \{0\}$: $\theta^0, \theta^1, \theta^2, \dots, \theta^{p^n-2}$ are all distinct and $\theta^{p^n-1} = \theta^0 = 1$. An indexing polynomial of degree n over $GF(p)$ is a monic irreducible polynomial whose zeros are primitive elements of $GF(p^n)$.

Alanen and Knuth searched for indexing polynomials by an algorithm based on the following:

THEOREM. $f(x) = x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_{n-1}x + a_n$ is an indexing polynomial over $GF(p)$ if and only if

T1. $(-1)^na_n$ is a primitive root of p ,

T2. the smallest positive integer r for which $x^r \equiv \text{integer} \pmod{f(x), \text{mod } p}$ is $r = (p^n - 1)/(p - 1)$ (in which case it can be shown that $x^r \equiv (-1)^na_n$).

3. ALANEN-KNUTH ALGORITHM

Alanen and Knuth's searching algorithm is as follows. Given p and n .

A0. [Initialize]

Factorization: $r = (p^n - 1)/(p - 1) = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}$, p_i are distinct primes and e_i are positive integers.

Set initial values: $a_1 = a_2 = \cdots = a_{n-1} = 0$ and $a_n =$ a primitive root of p .

A1. [Prime field element check]

If $f(\theta) \equiv 0 \pmod{p}$ for some $\theta \in GF(p)$ then go to A4.

A2. [r power check]

If $x^r \not\equiv (-1)^n a_n \pmod{f(x), \text{ mod } p}$ then $f(x)$ is not an indexing polynomial; go to A4.

A3. [Primitive check]

If $x^{r_i} \not\equiv$ integer $\pmod{f(x), \text{ mod } p}$ for every $r_i = r/p_i$ such that p_i is not a divisor of $p - 1$ then $f(x)$ is an indexing polynomial and the procedure terminates.

A4. [Loop]

Set $f(x)$ to be a new polynomial, with a_n fixed same; go to A1.

Step A2 is time consuming because almost all $f(x)$ which are tested will be found reducible.

4. NEW ALGORITHM

A speedup in the reducibility test is desirable. Berlekamp (1968) introduced a reducibility test as follows:

B. Let M be an $n \times n$ matrix whose i th row consists of the coefficients for the representation of $x^{p(i-1)} \pmod{f(x), \text{ mod } p}$. $f(x)$ is reducible if and only if $\text{rank}(M - I) < n - 1$, where I is the $n \times n$ identity matrix.

Our refined algorithm is as follows.

Given p and n .

A0. [Initialize]

A1. [Prime field check]

B0. [Reducibility check]

Construct matrix M defined in B.

If $\text{rank}(M - I) < n - 1$ then $f(x)$ is reducible; go to A4.

A2. [r power check]

A3. [Primitive check]

A4. [Loop]

TABLE I

Over $GF(3)$		$x^{19} + 2x^3 + 2x + 4$	$x^{11} + 15x + 18$
$x^{19} + x^2 + 2x + 1$		$x^{20} + 4x^2 + 3x + 3$	$x^{12} + 6x^2 + 3x + 5$
$x^{20} + x^5 + x + 2$		$x^{21} + 3x^2 + 3x + 4$	Over $GF(29)$
$x^{21} + x^3 + x + 1$		$x^{22} + x^3 + 3$	$x^7 + 2x + 27$
$x^{22} + x^3 + 2x^2 + 2x + 2$	Over $GF(11)$		$x^8 + x^2 + 8x + 2$
$x^{23} + x^3 + x + 1$	$x^9 + 2x + 9$		$x^9 + 3x + 27$
$x^{24} + 2x^4 + x^3 + 2x + 2$	$x^{10} + x^2 + 3x + 2$		$x^{10} + x^2 + 2x + 2$
$x^{25} + 2x^3 + 2x^2 + 2x + 1$	$x^{11} + 10x + 9$		$x^{11} + x^2 + 4x + 27$
$x^{26} + x^3 + x + 2$	$x^{12} + 4x + 2$		$x^{12} + 2x^2 + 9x + 2$
$x^{27} + x^5 + x^4 + 2x^3 + 2x + 1$	$x^{13} + 7x + 9$		Over $GF(31)$
$x^{28} + x^5 + x^4 + 2x + 2$	$x^{14} + 3x^2 + 3x + 2$		$x^7 + x + 28$
$x^{29} + x^4 + x^2 + 1$	$x^{15} + 2x^2 + 7x + 9$		$x^8 + 3x + 3$
$x^{30} + x + 2$	$x^{16} + 2x^2 + 3x + 2$		$x^9 + 23x + 28$
$x^{31} + x^3 + x + 1$	Over $GF(13)$		$x^{10} + 5x + 3$
$x^{32} + x^4 + x^3 + 2x + 2$	$x^9 + 7x + 11$		$x^{11} + 20x + 28$
$x^{33} + x^5 + x^4 + x^3 + 1$	$x^{10} + 2x^2 + 2x + 2$		$x^{12} + 3x^2 + 5x + 3$
$x^{34} + x^3 + 2x^2 + 2x + 2$	$x^{11} + 3x + 11$		Over $GF(37)$
$x^{35} + x^2 + 2x + 1$	$x^{12} + x^2 + x + 2$		$x^6 + 9x + 2$
$x^{36} + x^5 + x^4 + 2x^3 + 2$	$x^{13} + 12x + 11$		$x^7 + 7x + 35$
$x^{37} + x^3 + 2x^2 + 2x + 1$	$x^{14} + x + 2$		$x^8 + 15x + 2$
$x^{38} + x^3 + x^2 + 2x + 2$	$x^{15} + 2x^2 + 4x + 11$		$x^9 + x^2 + 5x + 35$
$x^{39} + x^5 + 2x^3 + 2x^2 + 1$	$x^{16} + x^3 + x^2 + 2x + 2$		$x^{10} + 14x + 2$
Over $GF(5)$	Over $GF(17)$		$x^{11} + 2x + 35$
$x^{13} + x^2 + 3x + 2$	$x^8 + x^2 + 3x + 3$	Over $GF(41)$	
$x^{14} + 2x^3 + x^2 + x + 3$	$x^9 + 7x + 14$	$x^6 + 3x + 6$	
$x^{15} + x^2 + 2$	$x^{10} + 6x + 3$	$x^7 + 6x + 35$	
$x^{16} + x^3 + 3x^2 + 3$	$x^{11} + 5x + 14$	$x^8 + x^2 + 9x + 6$	
$x^{17} + 2x^2 + 2x + 2$	$x^{12} + 7x^2 + 3x + 3$	$x^9 + 11x + 35$	
$x^{18} + x^4 + 3x^2 + x + 3$	$x^{13} + 15x + 14$	$x^{10} + x^2 + 14x + 6$	
$x^{19} + x^3 + 2x + 2$	Over $GF(19)$	$x^{11} + 20x + 35$	
$x^{20} + x^2 + 2x + 3$	$x^8 + x + 2$	Over $GF(43)$	
$x^{21} + 4x + 2$	$x^9 + x^2 + 2x + 17$	$x^6 + x^2 + 2x + 3$	
$x^{22} + x^2 + x + 3$	$x^{10} + 2x^2 + 6x + 2$	$x^7 + x^2 + 25x + 40$	
$x^{23} + 3x^2 + 2$	$x^{11} + 8x + 17$	$x^8 + 18x + 3$	
$x^{24} + 2x^4 + x^3 + 3x^2 + x + 3$	$x^{12} + 2x^2 + 2x + 2$	$x^9 + x^2 + 2x + 40$	
$x^{25} + 2x^3 + 3x + 2$	$x^{13} + 11x + 17$	$x^{10} + 9x + 3$	
$x^{26} + 2x^3 + x^2 + x + 3$	$x^{14} + x^2 + 3x + 2$	Over $GF(47)$	
Over $GF(7)$	Over $GF(23)$	$x^6 + x + 5$	
$x^{11} + x + 4$	$x^7 + 21x + 18$	$x^7 + 12x + 42$	
$x^{12} + 3x^2 + 2x + 3$	$x^8 + x^2 + 2x + 5$	$x^8 + 16x + 5$	
$x^{13} + x^2 + 5x + 4$	$x^9 + x + 18$	$x^9 + 7x + 42$	
$x^{14} + 2x^2 + 2x + 3$	$x^{10} + 7x + 5$	$x^{10} + 7x + 5$	
$x^{15} + 3x^2 + 3x + 4$		$x^{11} + 6x + 42$	
$x^{16} + 2x + 3$			
$x^{17} + x + 4$			
$x^{18} + 3x^3 + 3x + 3$			

Step B0 rejects nearly all the cases that would be rejected by A2. Aside from the time required for initialization A0, the time consumed in finding an indexing polynomial by the new algorithm is about

$$\frac{n^3/3 + 2n^2 + n}{(n^2 + 3n/2) \log_2 r} \times 100\%$$

of the old, which would be 65% for $p = 3$ and $n = 30$, 10% for $p = 23$ and $n = 10$.

The above expression is obtained easily by comparing the steps required for sweeping out and the step for A2 (Alanen and Knuth (1964, Appendix 2)).

5. RESULTS

Table I contains indexing polynomials of degree n over $GF(p)$, for $3 \leq p \leq 47$, $10^9 \leq p^n < 2^{62} \approx 10^{19}$.

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REFERENCES

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